

MOCH. ARIL INDRA PERMANA

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Portfolio◇ LinkedIn ◇ Github

SUMMARY

Visionary Senior Data Scientist and Technology Leader with a distinguished Physics background and 7+ years of pioneering excellence in artificial intelligence and machine learning innovation. Demonstrated expertise in architecting enterprise-scale AI solutions, leading cross-functional teams, and delivering transformative data-driven products that generate measurable business impact. Proven track record of founding and scaling technology ventures while mentoring the next generation of data professionals through comprehensive educational initiatives.

TECHNICAL SKILLS

Programming Languages & Frameworks:

- **Expert-Level:** Python
- **Advanced:** JavaScript (React, Node.js, Next.js), Java, C++
- **API Development:** FastAPI, Flask, RESTful Services

Artificial Intelligence & Machine Learning:

- **Deep Learning:** TensorFlow, PyTorch, Neural Networks, CNNs, RNNs
- **Generative AI:** Gemini, OpenAI GPT, Claude, LLM Fine-tuning, Prompt Engineering, RAG, Model Context Protocol
- **Computer Vision:** OpenCV, DocumentAI, OCR, Image Classification, Object Detection
- **NLP & Text Analytics:** NLTK, SpaCy, Transformers, Sentiment Analysis

Data Engineering & Analytics Platform:

- **Database Systems:** MySQL, BigQuery, MongoDB, ArangoDB, Elasticsearch, Redis
- **Big Data Ecosystem:** Apache Spark, Apache Airflow, BigQuery, Data Pipelines
- **Visualization & BI:** Metabase, Looker, Redash, Kibana, Tableau

Cloud Infrastructure & DevOps:

- **Google Cloud Platform:** VertexAI, Cloud Run, Cloud Storage, BigQuery, DocumentAI, Cloud SQL
- **DevOps & Orchestration:** Docker, Kubernetes, Git, CI/CD

PROFESSIONAL EXPERIENCE

Moskal.id

Co-Founder & Chief Technology Officer

Jun 2025 - Present

Jakarta

- Spearheaded the conceptualization and end-to-end development of an AI-powered social media intelligence platform, delivering real-time insights from multi-platform data streams (Twitter, Instagram, TikTok, News, Reddit, LinkedIn, Youtube, etc)
- Architected scalable, cloud-native infrastructure capable of processing millions of social media interactions daily, ensuring high availability and performance across global markets

- Pioneered advanced AI capabilities including Keyword Trends analysis, Topic Clustering, Key Opinion Leader (KOL) Monitoring, and Engagement Metrics with sophisticated filtering and search functionalities
- Developed intelligent AI Chatbot integrated with sentiment analysis and Generative AI for automated report generation and real-time user interactions
- Led product strategy and technology roadmap, defining go-to-market plans and aligning cross-functional teams with business objectives and technological vision
- Designed intuitive dashboard interface providing real-time metrics including mentions tracking, KOL analysis, and advanced data visualizations for enhanced decision-making
- Established robust data engineering pipelines utilizing NLP and machine learning algorithms for automated social media data collection, processing, and insight generation

PT Pakar Digital Global (Paper.id)

Jun 2023 - Present

Senior Data Scientist - AI Product Enablement

Jakarta

- Orchestrated enterprise-wide AI transformation initiatives, developing sophisticated machine learning models and analytics solutions that achieved 98.33% reduction in error detection time
- Revolutionized item categorization accuracy from 76% to 93% through advanced classification model optimization and feature engineering
- Pioneered comprehensive AI Product Enablement strategy, delivering end-to-end intelligent automation solutions for critical business operations
- Architected state-of-the-art OCR ecosystem for invoice and receipt processing, integrating DocumentAI, custom computer vision models, and advanced LLM prompting techniques
- Engineered sophisticated OCR annotation platform enabling streamlined data labeling workflows and continuous model improvement cycles
- Developed intelligent automated reconciliation systems for pre-disbursement and post-disbursement processes with intuitive platform interface, achieving 98% processing time reduction
- Implemented predictive Chart of Accounts (CoA) classification system leveraging NLP and machine learning for invoice item name analysis
- Created advanced conversational AI bot utilizing Model Context Protocol (MCP) and Retrieval-Augmented Generation (RAG) for enhanced user experience
- Built comprehensive KYC/KYB validation infrastructure featuring facial similarity scoring, document verification, and identity authentication
- Developed proprietary "askquinta" Python library for standardized error handling, automation, and operational efficiency
- Designed intelligent auto-selection system for Source of Fund (SOF) optimization in disbursement workflows
- Created executive-level dashboard for bank statement analysis powered by Gemini AI, reducing manual checking time by 98%

PT Bukalapak.com

Jan 2022 - Jun 2023

Assistant Manager - Audit Data Analytics

Jakarta

- Transformed audit operations through development of automated analytics dashboards using Redash and Looker, driving 70% improvement in team operational efficiency
- Pioneered sophisticated network analysis algorithms for fraud detection, successfully identifying complex syndicates and account farming schemes

- Architected intelligent automated response systems for audit data queries, eliminating manual intervention and accelerating decision-making processes
- Led high-impact fraud investigations utilizing advanced data science methodologies, uncovering critical internal and external fraudulent activities
- Executed comprehensive Quadrant Analysis for Marketing optimization, enhancing advertising partner selection strategies and ROI maximization
- Identified and resolved critical system vulnerabilities in Payment Gateway fee calculations, directly contributing to revenue enhancement and system integrity

PT Telkom Indonesia
Jan 2020 - Dec 2021
Data Scientist
Jakarta

- Engineered comprehensive legal analytics ecosystem processing data from 60+ legal websites, revolutionizing legal document analysis and compliance monitoring
- Developed advanced search engine and plagiarism detection system for legal document comparison using sophisticated NLP and similarity algorithms
- Implemented cutting-edge NLP models for legal regulation harmonization, streamlining compliance and regulatory analysis processes
- Conducted large-scale social media analytics achieving 80% user satisfaction in strategic decision-making processes
- Architected sentiment analysis models for real-time social media monitoring and brand reputation management
- Developed advanced topic clustering algorithms for automated content categorization and trend identification

PT Indonesia Indicator
Mar 2018 - Dec 2019
Data Scientist
Jakarta

- Pioneered innovative social scoring system for credit risk assessment leveraging social media data analytics and behavioral pattern recognition
- Developed automated reporting engine for comprehensive political campaign analysis and brand perception monitoring
- Conducted sophisticated network analysis for political trend identification and social media influence mapping
- Built intelligent bot-integrated reporting system enabling automated analytical report generation and insights delivery
- Implemented advanced sentiment analysis frameworks for real-time brand perception monitoring and reputation management

TEACHING & SPEAKING ENGAGEMENTS
Glints
2023
Data Science Education Leader
Online

- Delivered 14 high-impact educational webinars on advanced data science and AI methodologies to 500+ professionals
- Developed comprehensive curriculum encompassing Python programming, SQL fundamentals, machine learning, and advanced analytics
- Achieved exceptional 4.8/5 average satisfaction score, demonstrating outstanding content delivery and participant engagement

- Specialized topics: Python for Data Science, Advanced SQL, Pandas Data Manipulation, Linear Regression, and Predictive Modeling

Rumah Siap Kerja

Senior Data Analytics Instructor

2023

Online

- Led comprehensive 5-session data visualization masterclass for government-sponsored Prakerja professional development program
- Designed industry-standard curriculum covering data concepts, advanced analysis techniques, and visualization best practices
- Achieved 85% completion rate across multiple cohorts, significantly exceeding industry benchmarks
- Core modules: Data Analytics Fundamentals, Data Preparation, Advanced Analysis Tools, Strategic Visualization Techniques

Edspert

Excel Mastery Workshop Leader

2023

Online

- Conducted advanced Excel workshop focused on career advancement and professional productivity enhancement
- Achieved 90% participant satisfaction rate through hands-on learning and practical application methodologies
- Covered advanced Excel functions, PivotTables, data modeling, and professional visualization techniques

Pintarin AI

Data Science Evangelist

2023

Online

- Delivered comprehensive introduction to Data Science for emerging professionals and career changers
- Demonstrated practical applications using Python, Pandas, Scikit-learn, and industry-standard workflows
- Achieved 92% satisfaction score for content quality and delivery excellence

KEY PROJECTS

ECLIP (Electronic Clipping System)

Technical Lead & Architect

2023

Plabs

- Led development of enterprise-grade automated news monitoring system for West Java provincial government
- Architected sophisticated web scraping solution capable of monitoring 300+ news websites in real-time
- Implemented Elasticsearch-powered search engine with custom relevance ranking and semantic search capabilities
- Created comprehensive dashboard monitoring system using Kibana for real-time analytics and trend visualization
- Achieved 70% reduction in manual monitoring work through intelligent process automation and AI-driven categorization
- Integrated advanced sentiment analysis and trend tracking features for proactive media monitoring and crisis management

HONORS & AWARDS

- **Top 10 Finalist** - AI for Impact Hackathon APAC Region by Hack2skill (Dec 2024)

- Distinguished among elite innovators across Asia-Pacific for developing transformative AI solutions with measurable social impact
- Delivered cutting-edge AI applications utilizing Python, Generative AI, Gemini, VertexAI, DocumentAI, MLOps, Next.js, and Elasticsearch
- **Top 3 Winner** - IdeMenyala Hackathon Jakarta Innovation Challenge (Nov 2024)
 - Led cross-functional team to develop innovative smart city solutions addressing Jakarta's complex urban challenges
 - Implemented scalable solutions leveraging Python, FastAPI, Advanced Data Analytics, GCP, BigQuery, and IoT integration
- **Top 30 Finalist** - Telkomsel Tech Titans League Series 3 - Machine Learning Excellence Track (2021)
- **Top 30 Finalist** - Telkomsel Tech Titans League Series 1 - Data Science Innovation Track (2021)
- **Top 100 Finalist** - Telkomsel Tech Titans League Series 1 - Data Analytics Championship (2020)